Massey University
College of Sciences

161.304: Comments and Solutions for Computer Laboratory 4

Exercise 1: Three years of students' height

1. 4M4

   This is an more open-ended problem than the others. But perhaps the simplest model structure that addresses the prompt would be: (h height in cm, y year)

   $$\begin{aligned} h_i &\sim \text{Normal}(\mu, \sigma) \\ \mu &= \alpha + \beta y_i \\ \alpha &\sim \text{Normal}(150, 25) \\ \beta &\sim \text{Normal}(0, 10) \\ \sigma &\sim \text{Uniform}(0, 30) \end{aligned}$$

   Setting aside the issue of independence here - each student shows up in the data multiple times, thus making the time-series of heights dependent and potentially autocorrelated - no age range has been specified for our students in question. This leaves us with little information regarding potential priors (elementary school students are much smaller than university students). Therefore, I chose a weak prior with a large range on *alpha*.

   The prior on β is very weakly informative, centred on zero, which corresponds to no impact of year. This does not seem like a great prior, because surely height increases each year or stays the same. So maybe a Normal$(4, 1)$ prior? Uniform prior above zero would be better or a log-normal distribution? Truncated distribution would work too, but they haven't been introduced yet.

   The prior on σ will allow to cover a wide range of individual discrepancies.

   Notice that a linear modelling is enough, not need to go for higher-order polynomials over 3 years.

   If you feel strong, go and show me predictive checks :)

2. 4M5

   On the one hand, yes, I could use a prior that constraint β to be non-negative, e.g LogNormal$(2, 0.5)$ (plot it!).

   We may go further and observe that the growth probably excludes university students. So we could change our prior on α, Maybe on σ too if we have external information about ranges that make sense.

   On the other hand this is tricky, for the prior to really be "prior", it needs to be ignorance of the actual data values. So if this information was not scientific/common sense, but an observation in the data, basing the prior on it could get you into trouble. What kind of trouble? It could lead to an illusionary fit to data, as essentially you have used the data twice: once in setting the prior and once in conditioning on the data (computing the posterior). Much later in the book, you will see how constraints on the data can be used to design models, without running the risk of using the data twice.

3. 4M6

   OK the variance should in cm^2 :-/. So a $\sigma \sim$ Uniform$(0, 8)$ could do? Notice that we could revise the intercept too...if we knew the mean for sure, i.e. the age group.

Exercise 2: Refitting model 4.3; 4M7

```
library(rethinking)
data("Howell1")
d <- Howell1
d2 <- d[d$age >= 18,]
xbar <- mean(d2$weight)
m4.3U <- quap(alist(
    height ~ dnorm( mu, sigma),
    mu <- a + b * weight,
    a ~ dnorm( 178, 20),
    b ~ dlnorm( 0, 1),
    sigma ~ dunif(0, 50)
    ) ,
  data = d2
)
precis(m4.3U)
```

The estimates for β and σ are still the same, only α is changed. This makes sense since now in the uncentred version, the meaning of α has changed. Now, α is the average height for the case that the weight is 0. As there are no people with a weight of zero, this α is harder to interpret. We can compute μ in the uncentred version for when *weight* is the average weight. Unsurprisingly, we get the same value as the posterior mean for α in the centred model. The results for the two models are thus pretty much equal.

Let's check the covariance between parameters. Remember, in the centred version, the correlation between parameters was practically zero.

```
(vcm <- vcov(m4.3u))
```

We now observe some correlation between α and β. Let's check the correlation matrix:

```
cov2cor(vcm)
```

Now, there's a quite strong negative correlation between the two parameters.

What is happening here? Every time the slope parameter increases a bit, the intercept changes in the opposite direction, i.e. decreases.

You can check the posterior predictions for both models. And we do it then :)

```
weight.seq <- seq(from=25 , to=70 , by=1)
mu <- link(m4.3U, data=data.frame(weight = weight.seq))
# this is going to be for the regression line and uncertainty
mu.mean <- apply(mu, MARGIN = 2, mean)
mu.HPDI <- apply(mu, MARGIN = 2, HPDI)
# and this is for posterior predictions
sim.height <- sim(m4.3U , data=list(weight=weight.seq))
height.HPDI <- apply(sim.height, 2, HPDI, prob=0.89)
```

And we can do the plots, data first:

```
plot(height ~ weight , d2 , col=col.alpha(rangi2,0.5))
# then a posteriori mean line,
lines(weight.seq , mu.mean)
# HPDI region for line
```

```
shade(mu.HPDI, weight.seq)
# and HPDI region for simulated heights
shade(height.HPDI, weight.seq)
```

Exercise 3: More knots in the Cherry blossom spline and re-"priorise"; 4M7or8

First, let's load the data:

```
data("cherry_blossoms")
d <- cherry_blossoms
d2 <- d[ complete.cases(d$doy),]
```

Just to be sure to see an effect, I'm going to double the number of knots to 30:

```
library(splines)
num_knots <- 30
knot_list <- quantile(d2$year, probs = seq(0, 1, length.out = num_knots))

B <- bs(d2$year,
  knots=knot_list[-c(1, num_knots)],
  degree=3, intercept=TRUE)

ms <- quap(alist(
   D ~ dnorm(mu, sigma),
   mu <- a + B %*% w,
   a ~ dnorm(100, 10),
   w ~ dnorm(0, 10),
   sigma ~ dexp(1)
 ), data=list(D=d2$doy, B=B),
 start = list(w=rep(0, ncol(B)))
)
```

The curve with 30 knots is less smooth and even wigglier than the curve with 15 knots.

Let's now also change the width of the prior on the weights. I am going to change the standard deviation for the prior on the weights w to 100 (before it was 10).

Do it!! Use the R code blocks 4.77 and 4.78, p119. And tell me if you agree with the remarks below.

It got even wigglier! This is a bit difficult to see because the curve with 15 knots and the narrower priors was quite wiggly but it can for example be seen at the end to the right.

More knots means there are more basis functions and so the curve can fit smaller local details. Fitting only a handful of knots will force the model to fit the more global trends: try with 5 knots!

If we turn up the standard deviation of the prior we allow the weights to become larger. If the standard deviation is small, the weights will be closer to 0 and thus wiggle around closer to the mean line. If the weights become larger, we allow the curve to have more peaks. Do play around!

Exercise 4: More Cherry Blossom

1. 4H5

   First, let's plot the two variables.

```
d <- cherry_blossoms
plot(doy ~ temp, d, col=col.alpha(rangi2, 0.5), pch=20, cex=1.4,
      ylab = "Day of Year", xlab="Temperature [°C]")
mtext("Temperature vs Day of Blossom")
```

It looks like there is a negative relationship between the two: the higher the temperature, the earlier in the year does it start to blossom.

Since there are many missing values, I'll restrict the data to cases where both `doy` and `temp` exist.

First, I'll try a linear model:

```
mean_temp <- mean(d2$temp)
ml <- quap(alist(
    D ~ dnorm(mu, sigma),
    mu <- a + b*(temp - mean_temp),
    a ~ dnorm(100, 10),
    b ~ dnorm(0, 1),
    sigma ~ dexp(1)
  ), data=list(D=d2$doy, temp=d2$temp)
)
precis(ml)
```

Indeed, the model estimates a negative slope β. You can plot the model: posterior on the lines and posterior predictive distribution. The temperature does have some effect on the day of first blossom but note that the uncertainty is quite high.

I will then try both a quadratic and a cubic model. Quadratic first:

```
mquadratic <- quap(alist(
    D ~ dnorm(mu, sigma),
    mu <- a + b1*temp_s + b2*temp_s2,
    a ~ dnorm(100, 10),
    b1 ~ dnorm(0, 1),
    b2  ~ dnorm(0, 1),
    sigma ~ dexp(1)
  ), data=list(D=d2$doy, temp_s=d2$temp_s, temp_s2 = d2$temp_s2)
)
```

And cubic:

```
mcubic <- quap(
  alist(
    D ~ dnorm(mu, sigma),
    mu <- a + b1*temp_s + b2*temp_s2 + b3*temp_s3,
    a ~ dnorm(100, 10),
    b1 ~ dnorm(0, 1),
    b2 ~ dnorm(0, 1),
    b3 ~ dnorm(0, 1),
    sigma ~ dexp(1)
  ), data=list(D=d2$doy, temp_s=d2$temp_s, temp_s2 = d2$temp_s2,
    temp_s3 = d2$temp_s3)
)
```

Interestingly, the two models look pretty much the same as the linear model. Check it on the posterior of the parabolas and cubic lines and on the posteior predictive distributions! So there is no gain anything in using a polynomial model, instead we only loose interpretability in the parameters with the quadratic and cubic terms left with very small values.

Lastly, let's also check out how splines perform for this example. I will use 15 knots and the same priors as also used in the chapter on the weight.

```
num_knots <- 15
knot_list <- quantile(d2$temp, probs = seq(0, 1, length.out = num_knots))

B <- bs(d2$temp,
   knots=knot_list[-c(1, num_knots)],
   degree=3, intercept=TRUE)

ms <- quap(alist(
    D ~ dnorm(mu, sigma),
    mu <- a + B %*% w,
    a ~ dnorm(100, 10),
    w ~ dnorm(0, 10),
    sigma ~ dexp(1)
  ), data=list(D=d2$doy, B=B) ,
  start = list( w=rep( 0, ncol(B)))
)
```

The splines are wigglier and have one especially weird dip just for temperatures below 6degC. Personally, I am sceptical this dip makes physically sense and it isn't just an artefact of the splines being too wiggly. Comparing the three models, in this case, I would go with the simple linear model.

2. 4H6

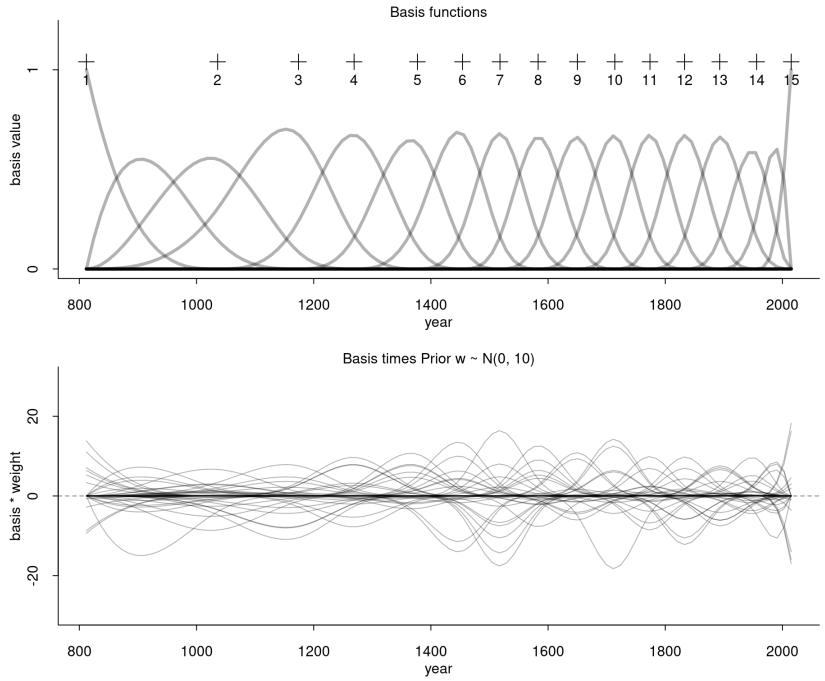
   I will only draw 10 different samples for each weight so that the plot won't get too cluttered.

```
num_knots <- 15
knot_list <- quantile(d2$year, probs = seq(0, 1, length.out = num_knots))
years <- seq(from=min(d2$year), to=max(d2$year), length.out = 100)

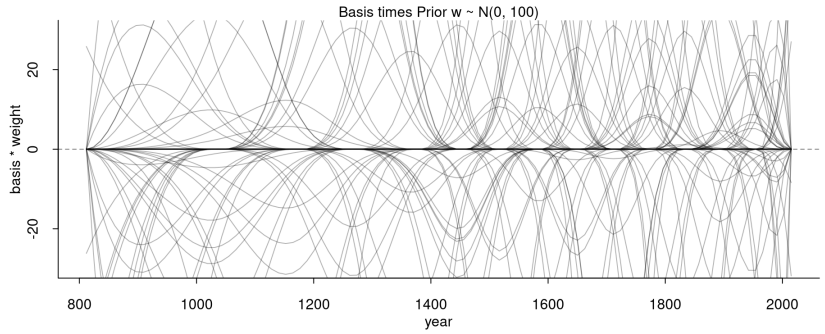
B <- bs(years, knots=knot_list[-c(1, num_knots)],
     degree=3, intercept=TRUE)

N <- 10
w <- rnorm(N*ncol(B), 0, 10)
w <- matrix(w, nrow=ncol(B))
```

In the top, we have again the basis functions and in the bottom we see the basis functions multiplied with weight values samples from our prior:



Compare this to the curves we get for the wider prior:



With this wider prior, the curve go much higher and lower. With the standard deviation of the prior we can thus control how high or low the curve can wiggle away from the mean.

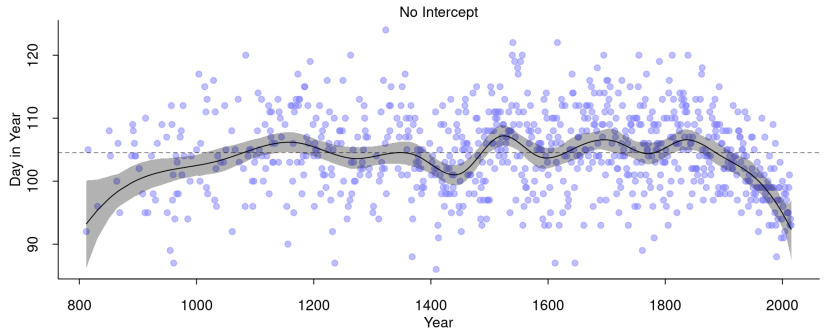
3. 4H7 (labelled 4H8?)

   Let's see what happens if we just omit the intercept:

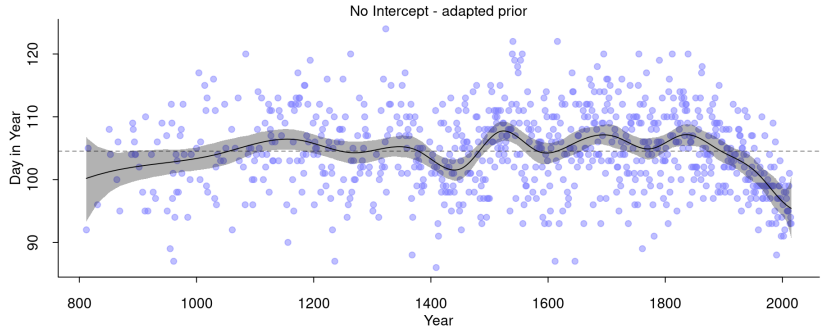
```
num_knots <- 15
knot_list <- quantile(d2$year, probs = seq(0, 1, length.out = num_knots))
B <- bs(d2$year, knots=knot_list[-c(1, num_knots)],
        degree=3, intercept=TRUE)

m <- quap(alist(
    D ~ dnorm(mu, sigma),
    mu <- B %*% w,    # removed intercept
    w ~ dnorm(0, 10),
    sigma ~ dexp(10)
  ), data=list(D=d2$doy, B=B) ,
  start = list( w=rep( 0, ncol(B)))
)
```

   Let's see the result:



This almost worked, except that we now have that the splines seem to come from far down to the left. This is probably since we specified that the weights are distributed around 0: $w \sim \text{Normal}(0, 10)$. Let's fix this by changing the prior on w to this: $\text{Normal}(100, 10)$. This is basically the prior we before had on the intercept.



Now, we basically got the same model as before that included the intercept.